Landscape Fragmentation and Conservation Strategies for Cheetahs in Northern South Africa

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Abstract

Rapid habitat fragmentation and escalating human-wildlife conflict have contributed to steep global declines in cheetah (*Acinonyx jubatus*) populations, currently estimated at approximately 7,500 individuals worldwide (Weise et al., 2015). South Africa holds roughly 1,000 of these individuals and represents the only national population currently exhibiting net growth (Marnewick et al., 2014). This study analyzes how the landscape of the Limpopo province in northern South Africa can be leveraged to support expanded cheetah populations using three complementary approaches: NDVI reclassification of satellite-derived remote sensing data to assess vegetation suitability; multi-criteria analysis (MCDA) to evaluate tracking method effectiveness; and effective population size estimation to contextualize minimum viable population (MVP) requirements. Approximately 72% of the free-range study area falls within vegetation classes broadly suitable for cheetah occupancy (Classes 2–3: grassland and mixed open woodland). MVP analysis yielded an estimated effective population size of $N_e \approx 984$, approaching the revised minimum threshold of 1,000 recommended by Frankham et al. (2014). GPS collar tracking and camera trapping emerged as the highest-rated monitoring methods under regional logis-

tical constraints. These findings underscore the urgency of landscape-scale conservation strategies that extend beyond formally protected area boundaries.

Keywords: cheetah, *Acinonyx jubatus*, habitat fragmentation, minimum viable population, NDVI, remote sensing, South Africa

Introduction

The cheetah (*Acinonyx jubatus*) occupies an increasingly precarious position among African large carnivores. Once broadly distributed across Africa, the Middle East, and Central Asia, the species has been extirpated from the majority of its historical range and is now most abundant in four southern African countries: Namibia, Botswana, South Africa, and Zimbabwe (Weise et al., 2015). Conservative global population estimates of approximately 7,500 individuals continue to decline across most range states (Durant et al., 2017). The species faces a convergence of threats: habitat loss and fragmentation, depletion of prey populations, persistent human-wildlife conflict, and the challenge of maintaining viable populations predominantly outside formally protected areas (Weise et al., 2015).

Within South Africa, the Limpopo province represents a critical range state. The landscape is characterized by extensive smallholder agriculture interspersed with private game reserves and pockets of conservation land. Cheetahs in this region are embedded within a working landscape in which the majority of individuals occupy unprotected land and regularly encounter livestock operations, a context that generates persistent predator-farmer conflict and complicates systematic population monitoring.

The present study examines three interconnected dimensions of cheetah conservation in northern South Africa. First, vegetative suitability is characterized using NDVI reclassification to identify areas consistent with documented cheetah habitat requirements. Second, available monitoring methods are evaluated using MCDA to assess relative effectiveness under the logistical constraints of the study region. Third, a minimum viable population estimate is derived to situate the current global population relative to long-term viability thresholds. Together, these analyses are intended to inform strategic prioritization of conservation effort in a region where targeted intervention can have disproportionate global impact.

Study Region

The study region encompasses the Limpopo province of South Africa, delimited to an area of approximately 50,000 km^2. The terrain is predominantly arid and flat, with intermittent rocky outcroppings and low mountain ranges. Vegetation is characteristic of the Savanna Biome, transitioning between grassland, bushveld, and mixed woodland, conditions consistent with documented cheetah habitat preferences for open to semi-open landscapes.

Cheetahs in this landscape maintain large, partially overlapping home ranges. Female home ranges average approximately 650–800 km^2, while male territories span approximately 39–78 km^2; non-territorial individuals referred to as "floaters" have been recorded at ranges up to 1,600 km^2 (Melzheimer et al., 2020). A range-wide survey found that within 400,000 km^2 surveyed across southern Africa, approximately 55% of the known cheetah population occupied largely unprotected land (Weise et al., 2015). When full land-use overlap with human activity is incorporated, this proportion exceeds 90%, underscoring the need for conservation frameworks that operate beyond formally designated reserve boundaries.

Prior habitat use research established that cheetahs preferentially select areas with 25–50% woody cover and grasses of medium height, consistent with a need for both concealment during stalking and unobstructed space for high-speed pursuit (Rejmanek et al., 2013). A discriminant function analysis correctly classified 72.1% of "used" habitats and 70.4% of "non-use" habitats based on these structural vegetation characteristics (Rejmanek et al., 2013).

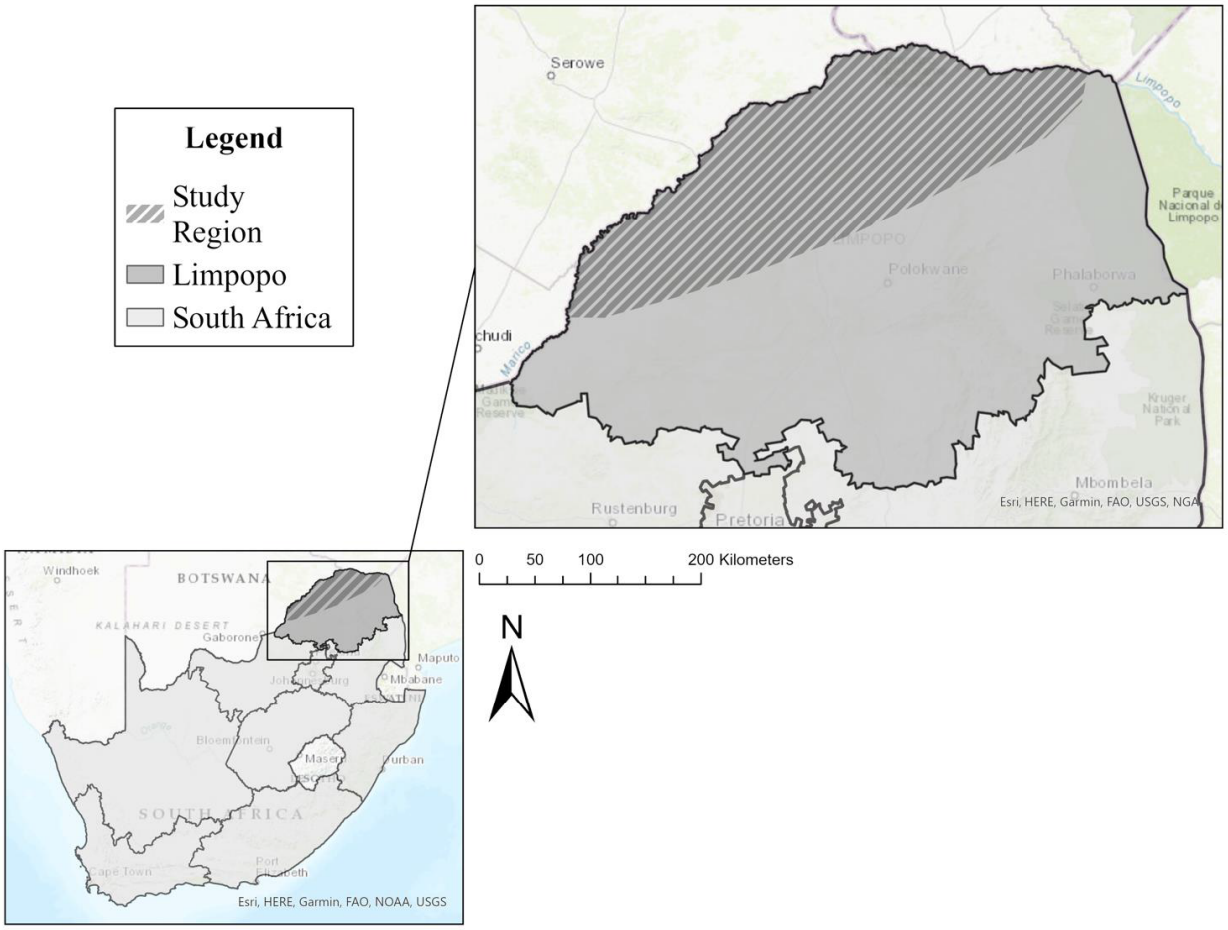

Figure 1: *Area of Interest for This Study With Callout for Reference Within South Africa.* The hatched region indicates the study area within the Limpopo province. The inset map (lower left) shows the location of Limpopo within the broader southern African context.

Cheetah Behavioral Ecology

Cheetah spatial ecology is organized around a hierarchical system of territory, range use, and transient movement. Sexual maturity is reached at approximately two to three years of age. Females are not territorial and range broadly across overlapping home ranges, practicing polyandry, a reproductive strategy whereby individual litters may be sired by multiple males, supporting genetic diversity within isolated populations (Melzheimer et al., 2020).

Male spatial organization is more structured. Territory-holding males, typically forming coalitions, maintain defined home ranges and aggregate at specific landscape fea-

tures termed "communication hubs" (CHs), spatial nodes identified through GPS tracking data at which conspecific encounter rates are elevated (Melzheimer et al., 2020). CHs are of particular conservation relevance because they disproportionately overlap with livestock grazing areas, making them the primary locus of human-cheetah conflict in the region. Non-coalition males range more broadly and contribute less directly to the breeding population but represent an important reservoir of genetic diversity.

This spatial framework is fundamental to monitoring design. Population censuses that do not account for the heterogeneous distribution of cheetah space use, with concentrations at CHs alongside diffuse occupancy across the broader landscape, will systematically misrepresent population density and distribution.

Tracking Methods

Effective population monitoring is a prerequisite for evidence-based conservation planning, yet no single method has proven both logistically feasible and sufficiently comprehensive for cheetah assessment across large, heterogeneous landscapes. Conventional approaches include point count transects, camera trap arrays, and GPS collar tracking. An emerging alternative, morphometric footprint identification, has demonstrated promise as a low-cost, non-invasive tool (Baralle et al., 2021).

The logistical demands of conventional methods are substantial when scaled to the study area. Point count surveys require researchers to walk approximately 16 km per 100 km^2 grid cell to achieve 95% confidence of species absence. Covering the full 50,000 km^2 study area would require approximately 8,000 km of survey effort, equivalent to roughly 1,500 person-hours. Camera trapping requires 193 trap-nights per 100 km^2 cell for equivalent confidence; at full study area scale, this corresponds to deploying 500 cameras over 193 nights, with equipment costs exceeding $30,000 at $60 per unit, before accounting for labor or landowner permissions.

These constraints illustrate that the spatial scale of viable cheetah habitat in northern South Africa exceeds the capacity of any single monitoring approach. A coordinated,

multi-method strategy is therefore not merely advisable but operationally necessary.

Data Analysis

Multi-Criteria Analysis of Tracking Methods

A multi-criteria analysis (MCDA) was conducted to evaluate the relative effectiveness of four monitoring approaches: morphometric footprint identification, GPS collar tracking, camera trapping, and point count surveys. Nine attributes were assessed: technology maturity (T), spatial coverage (C), accuracy (A), data transmission capacity (DT), battery life or persistence (BL), cost (Co), data storage requirements (DS), range of applications (Ap), and implementation challenges (Ch). Each attribute was assigned a weight reflecting its relative importance to regional monitoring objectives, and each method was scored on a 1–5 scale. Weighted scores were summed to produce an overall effectiveness rating (Table 1).

Table 1
Multi-Criteria Analysis Scores and Weighted Totals for Cheetah Monitoring Methods

Attribute	**Wt.**	**Morphometry**	**GPS Collars**	**Camera Traps**	**Point Counts**
Technology (T)	.15	4	3	3	1
Coverage (C)	.20	5	4	2	1
Accuracy (A)	.20	4	3	3	4
Data Transmission (DT)	.10	5	4	5	5
Battery Life (BL)	.05	2	4	5	5
Cost (Co)	.10	1	3	5	5
Data Storage (DS)	.05	3	4	3	4
Applications (Ap)	.10	4	3	5	4
Challenges (Ch)	.05	3	3	2	2
Weighted Total		**3.90**	**3.45**	**3.40**	**2.85**

Note. Wt. = attribute weight. Scores reflect performance on a 1 (low) to 5 (high) scale. Weighted totals are the sum of (score × weight) across all nine criteria.

Morphometry scored highest overall (3.90), driven by superior coverage potential and broad applicability, though its current operational maturity remains limited. GPS collar tracking (3.45) and camera trapping (3.40) represent the most practically effective conventional methods. Point counts (2.85) ranked lowest and are susceptible to substantial human error; their utility is best realized as a supplementary gap-filling approach rather than a primary census tool.

Vegetation Analysis

Spatial data preparation.. Primary geospatial data sources included national and provincial boundary shapefiles for South Africa, supplemented by study-specific shapefiles delineating the study region, protected land, and free-range land. Free-range area was derived by erasing conservation and protected land polygons from the full Limpopo provincial boundary using overlay analysis in ArcGIS (Esri). The resulting free-range shapefile was clipped to the study region extent. NDVI raster data were acquired for the full Limpopo province to allow regional vegetation patterns to contextualize conditions within the study area. Data sources included GEDI L3 Gridded Land Surface Metrics accessed via EarthExplorer and Open Topography.

NDVI reclassification.. The continuous NDVI raster was reclassified into five discrete vegetation classes representing distinct biome types (Table 2). Class boundaries were defined based on the empirical distribution of pixel values across the study region and calibrated against documented vegetation associations relevant to cheetah habitat use. Symbology followed standard NDVI convention (red to green). Zonal statistics were calculated to quantify the proportional area of each class within both the full Limpopo province and the free-range study region.

Table 2
NDVI Reclassification Scheme and Associated Vegetation Biome Descriptions

Class	NDVI Range	Vegetation Description
1	122–1327	Sparse vegetation/bare ground; desert and rocky terrain
2	1327–1949	Grasslands and shrublands; moderate vegetation cover
3	1949–2494	Mixed vegetation; sparse woodland and transitional zones
4	2494–3116	Dense woodland; well-established forest
5	3116–10037	High-productivity vegetation; wetland and agricultural areas

Note. Class boundaries were defined based on the distribution of raw NDVI pixel values and calibrated to recognized southern African vegetation biome classifications.

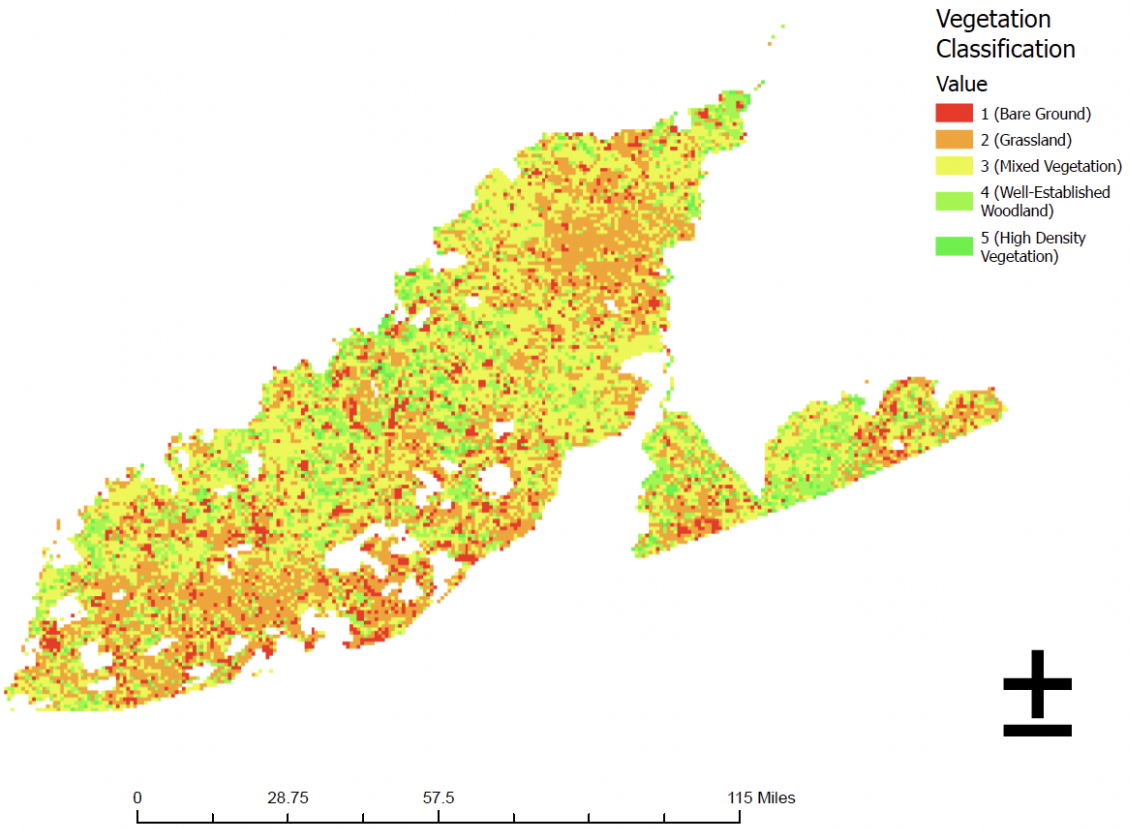

Figure 2: *Free-Range Land Within the Study Region Overlaid With Reclassified NDVI Vegetation Values.* Class 1 (bare ground) is shown in red; Class 2 (grassland) in orange; Class 3 (mixed vegetation) in yellow; Class 4 (well-established woodland) in light green; Class 5 (high-density vegetation) in dark green. Scale bar in miles.

Results

Vegetation Cover Analysis

Across the full Limpopo province, Class 2 (grasslands/shrublands) and Class 3 (mixed/low- density vegetation) collectively dominated the landscape, comprising 31.24% and 30.39% of total area, respectively. Class 4 (dense woodland) accounted for 18.17%, while Class 1 (sparse/bare ground) and Class 5 (high-productivity zones) comprised 11.55% and 8.64%.

Within the free-range portion of the study region, the dominance of lower-density

vegetation classes was more pronounced. Class 3 comprised 40.05% of the area, followed by Class 2 at 32.00%. Class 4 covered 16.71%, with Class 1 (8.48%) and Class 5 (2.75%) comprising smaller fractions.

Taken together, approximately 72% of the free-range study area falls within Classes 2 and 3, the vegetation types most consistent with documented cheetah habitat preferences (Rejmanek et al., 2013). Dense woodland (Class 4) and high-productivity vegetation (Class 5) are generally less suitable, as they reduce visibility and constrain high-speed pursuit. Sparse bare-ground areas (Class 1) similarly offer limited value due to reduced prey availability and cover. These results suggest that the Limpopo free-range landscape retains substantial ecological capacity to support cheetah populations, contingent on effective management of fragmentation and human-use pressures.

Minimum Viable Population Analysis

The minimum viable population (MVP) concept defines the smallest population size required to sustain long-term demographic and genetic viability, accounting for environmental stochasticity, demographic variation, genetic erosion, and catastrophic risk. A formal cheetah-specific MVP has not been established in the peer-reviewed literature. Drawing on the global population estimate of Durant et al. (2017) and applying the effective population size (N_e) framework used for analogous large carnivore assessments, an approximation was derived.

Of an estimated global population of 7,100 wild cheetahs, approximately 40–50% are estimated to be pre-reproductive juveniles. Assuming a conservative adult fraction of 60%, approximately 4,260 adults are reproductively eligible. A 50:50 sex ratio yields 2,130 adult females and 2,130 adult males. Male reproductive access is heavily mediated by coalition membership and spatial proximity to communication hubs (Melzheimer et al., 2020); approximately 15% of adult males are estimated to be actively breeding. Females were estimated at 50% active breeding participation. This yields a breeding female population ($N_f \approx 1{,}065$) and a breeding male population ($N_m \approx 320$).

Effective population size was calculated using the standard formula for populations with unequal sex ratios:

$$N_e = \frac{4 \cdot N_m \cdot N_f}{N_m + N_f} \tag{1}$$

$$N_e = \frac{4 \cdot 320 \cdot 1{,}065}{320 + 1{,}065} \approx 984 \tag{2}$$

This estimate places the current effective cheetah population just below the revised minimum of $N_e \geq 1{,}000$ recommended by Frankham et al. (2014) to limit inbreeding depression and fitness loss to $\leq$10% over five generations. The severity of the underlying genetic risk is underscored by evidence that global cheetah populations were once reduced to only a few hundred individuals, a bottleneck sufficient to compress genetic diversity to levels approaching that of identical twins (Durant et al., 2017). The current population thus occupies a critical threshold: large enough to sustain incremental recovery under favorable conditions, but sufficiently constrained that continued habitat loss or elevated conflict-driven mortality could push N_e below the long-term viability threshold.

Conclusion

This study presents a spatially grounded assessment of cheetah conservation viability in the Limpopo province of northern South Africa, integrating vegetation habitat analysis, monitoring method evaluation, and population viability estimation. The results collectively indicate that the region retains meaningful ecological potential for cheetah conservation, while identifying structural vulnerabilities that constrain that potential.

The NDVI analysis demonstrates that approximately 72% of free-range land in the study area supports vegetation conditions consistent with cheetah habitat requirements. However, the utility of this habitat depends on landscape connectivity. Fragmentation by roads, fencing, and agricultural development disrupts corridor function and reduces effective range even where vegetation structure remains appropriate (Durant et al., 2017).

Future analyses should incorporate landscape permeability modeling, incorporating road buffers, land tenure boundaries, and prey availability layers, to advance from habitat suitability characterization toward functional corridor planning.

The MVP analysis underscores the fragility of the current global cheetah population. An estimated N_e of approximately 984 places the species at the lower bound of long-term genetic viability as defined by Frankham et al. (2014), and the species' documented historical bottleneck confirms that this risk is not theoretical (Durant et al., 2017). Maintaining population connectivity across the fragmented southern African landscape is therefore both a spatial management priority and a conservation genetics imperative.

South Africa's demonstrated capacity to recover its cheetah population from fewer than 300 individuals in the early 2000s to approximately 1,000 today provides evidence that recovery is achievable under targeted intervention (Marnewick et al., 2014). Scaling this success requires refined habitat models informed by fine-resolution vegetation structure data (e.g., space-based LiDAR), GPS collar movement records for ground-truthing habitat use, and prey density modeling. Integrating these data streams will substantially improve precision in conservation targeting and strengthen the evidentiary basis for landscape-scale management decisions.

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